



Using Augmented Video to Test In-Car User Experiences of Context Analog HUDs

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Abstract

Automotive interactive technologies represent an exemplar challenge for user experience (UX) designers, as the concerns for aesthetics, functionality and usability add up to the compelling issues of safety and cognitive demand. This extended abstract presents a methodology for the user-centred creation and evaluation of novel in-car applications, involving real users in realistic use settings. As a case study, we present the methodologies of an ideation workshop in a simulated environment and the evaluation of six design idea prototypes for in-vehicle head up display (HUD) applications using a semi-naturalistic drive. Both methods rely on video recordings of real traffic situations that the users are familiar with and/or experienced themselves. The extended abstract presents experiences and results from the evaluation and reflection on our methods.

Author Keywords

Automotive user interfaces, user centred design, user experience, head up display

ACM Classification Keywords

H.4.3 [Information Systems Applications]:
Communications Applications; H.5.2 [Information
Interfaces and Presentation (e.g., HCI)]: User Interfaces

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AutomotiveUI '14, Sep 17-19 2014, Seattle, WA, USA
ACM 978-1-4503-0725-3/14/09.
<http://dx.doi.org/10.1145/2667239.2667302>

Introduction

Interaction design in general and User Experience (UX) evaluation in particular are heavily based on user testing and rely on early feedback from real users in a real(istic) use setting. UX captures the hedonic and aesthetic dimensions of interactive technologies from a subjective perspective and in a highly dynamic and concrete situation [3]; as such, not surprisingly, the advantage of field studies versus laboratory experiments has often been acknowledged (see e.g. [10]). The development of methods to evaluate concept ideas and non-functional prototypes has been indicated as a priority in UX research [11].

Despite this importance, there are contexts where the evaluation of early prototypes in the field is not practical or feasible. Reasons for this include the additional cost and effort of field studies, the lack of control on the experimental conditions, the difficulty of testing early prototypes that may only be partially functional, and lastly, the possible additional safety risk for participants. The additional risk for participants is particularly apparent in the context of designing user/driver experiences, and in fact a large majority of studies that investigate interactive technologies in the car rely on laboratory experiments [5].

UX research in the car used to be limited to the in-car entertainment system (and the actual design of the car in a broader UX sense), which was then expanded to other mundane tertiary tasks such as making phone calls, operating a navigation system, or, more recently, exploring points of interest. Recent technology advancements, however, provide new opportunities to create novel and vastly different in-car user experiences. These advancements are made a) in the availability of place- and time-specific digital information, and b) in hardware that

allows us to access or display this information.

In this extended abstract, our primary focus is on head-up displays (HUDs) and their ability to visually represent digital information. HUDs are not only available in luxury cars, but are also starting to make their way into the car after market, such as the Navdy or Pioneer Cyber Navi. In addition, the Auto-UI community continues to innovate and refine HUDs towards contact-analog HUDs [6].

Contact-analog HUDs draw from augmented reality research. They allow digital information to be rendered and spatially related to elements of the real-world scenery within the field of view of the driver [6].

Examples applications tend to focus on the safety applications, such as highlighting pedestrians in bad weather conditions. Others include attempts to influencing social norms and stimulate a greener driving style, mediating and amplifying social cues and regulating over/under stimulation in young drivers [7]; providing adaptive feedback on lane adherence to improve driver behavior [1]; provide navigation information specially tailored to elder drivers [4], and more.

This extended abstract presents a novel methodology for the exploration, creation and early evaluation of the UX of in-car HUD applications.

Our Methodological Aims

The main question that drives the exploration is: *What data/information or applications can offer new user experiences to future drivers?* We argue that the advancements in mobile and Auto-UI technologies over the past years turn this into a timely question. Furthermore, revisiting this question freely, without safety constraints (e.g. driver distraction), may unlock



Figure 1: Advanced driving simulator.

innovative ideas that might retrospectively be designed towards having potential safety benefits, therefore uniting the user experience and road safety, rather than seeing them as two contradicting concepts.

The aim of the two methods presented in this extended abstract was to a) generate as many ideas as possible without any constraints, and b) allow for an early, quick evaluation of those ideas.

Firstly, a guided ideation (idea creation) workshop in a simulated environment was adopted to inspire the novel design ideas we sought. Then, a semi-naturalistic driving experience was used to evaluate their potential to improve the user experience and have safety benefits.

Being able to evaluate early, low fidelity prototypes can help designers to focus on those ideas that work, long before addressing potential distraction. For this to be possible, the driving context must be re-created in a safer, possibly controlled, testing environment.

Driving simulation is a valuable tool for evaluating attention, reaction times and driving accuracy, but the validity of simulation studies has been questioned when it comes to assessing the effects of complex tasks involving social interaction, even for something as simple as a phone call [2]. In fact, to study the User Experience of automotive user interfaces, methods and frameworks for fostering creativity and supporting the evaluation of early prototypes are to be considered a compelling priority.

Guided Ideation (Idea Creation) Workshop in a Simulated Environment

A guided ideation workshop was the starting point of our exploration. We invited 14 expert users and early adopters of mobile technologies and location-based services. They

were recruited through the university's research labs in those areas. The participants were then grouped into seven groups of two so that they would know and be familiar with other. The seven groups then participated in seven guided, 1-hour ideation (idea creation) workshops inside an advanced driving simulator. The workshop was designed to a) create an immersive and realistic driving scenario, b) be conducive to a collaborative, creative brainstorming, and of course c) be safe.

The advanced simulator provides a complete Holden Commodore vehicle mounted on a six degrees of freedom (6DOF) motion platform and 180 degree projection screen. The simulator is generally operated in conjunction with a simulation software (SCANeR studio), see Fig. 1. However, we decided to display real-world driving videos on the projection screen instead, because we assumed that the many details of the real-world would inspire more ideas than the simplified world provided by the simulation software.

The videos were specifically recorded for this purpose and featured various driving scenarios, such as inner city, inner-city bypass and highway routes that would have been familiar to the participants, as well as typical suburban and scenic routes of the surrounding area. It is important to note that participants were not asked to drive the simulator, they were only asked to watch the videos through the front windscreen while sitting in a real car cabin, and immerse and imagine themselves in the driving scenarios presented by the videos (e.g., going grocery shopping in the suburbs, driving home from work through the inner city, a weekend trip with the family on the highway). The primary goal was to allow participants to focus on the brainstorming of novel applications specific to each of the presented driving contexts.

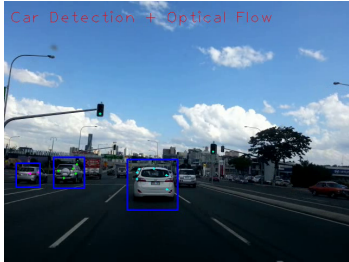


Figure 2: Automatic processing of video footage from a naturalistic drive.



Figure 3: Augmented video of a naturalistic driving session.



Figure 4: In-lab evaluation of design ideas using augmented video.

One of the two participants was seated in the drivers seat, the other participant was seated in the front passenger seat. One researcher acted as the facilitator seated on the backseat, while a research assistant operated the simulator and supported the researcher through note taking.

The facilitator gave a brief introduction to each of the driving scenarios, followed by questions and tasks that guided the ideation process, such as the following: If there would be *no constraints* about what is technically possible and safe, what kind of information (data, text, images, video, sound, etc.) would you personally like to be conveyed to you in these different driving contexts? And from your professional perspective, what kind of data could you envisage to be conveyed to future drivers? We considered freeing participants from all constraints as another important ingredient to inspire ideas. Lastly, participants received whiteboard markers to sketch their ideas on the car's windscreen if they wanted to represent their ideas visually.

The ideas were recorded and grouped into themes. The results were then presented back to 10 of the previous participants in a 2-hour focus group two weeks after the initial ideation workshop for further in-depth discussions. The focus group was facilitated by the same researcher as the ideation workshop and again assisted through note taking by another researcher.

A small subset of the generated ideas, which we thought could be implemented for in-vehicle HUD applications to potentially have safety benefits (in this case mitigating aggressive driving), were then evaluated early using a similar real-world video methodology, which is discussed in detail in the following section.

Semi-Naturalistic Drive

Early evaluation of research prototypes was performed using a semi-naturalistic drive. First, 11 participants were invited to drive in a naturalistic fashion in real traffic conditions. Each drive was video recorded and specific driving events were identified. We designed software that processed the video footage in real time to identify relevant elements, and add graphics to the participants' personally experienced driving scenes. This simulated the intended design ideas and presented the participants with contextualized simulations of the future technologies. Figures 2, 3 and 4 describe the process.

This augmented video footage was used to facilitate a semi-structured interview only a few minutes after the drive, when the memory of the real drive and associated emotions were still fresh. During the interview, participants provided retrospective evaluations about certain aspects of the driving experience, their corresponding feelings, cognition and behavior. In addition, they were able to evaluate the visual HUD prototypes in light of experiencing their use as they would have been used in the real driving context.

The idea of this method was to retain part of the realism of a naturalistic drive, as the video presented was taken from the real life of the research participant, rather than from a standard collection or computer generated. Driving events, interactions, conflicts, could be easily matched to the real emotional and cognitive appraisal, specially because the video was presented short after the naturalistic drive.

Discussion

The ideation workshop generated a rich amount of 80 unique design ideas for various different driving scenarios.

This result, in our opinion, validated the method as being highly conducive to the creation of new ideas. Particularly having two participants and one facilitator in the car worked very well. As expected from a brainstorming session, the two participants would often nudge each other towards the creation of a new idea. In those sessions where participants were initially slow to get started, the facilitator was able to nudge them towards ideas by reusing some of the ideas from previous sessions. Having only two participants in the car allowed each of them adequate time to comfortably communicate their ideas. The "privacy" of the car also seemed to contribute to the participants' comfort. Lastly, participants intrinsically tended to constrain themselves by considering safety aspects of their ideas (presumably triggered by the realistic immersion of the method), which required the facilitator to constantly push them towards a free exploration of opportunities rather than the safety issues.

Eventually the method led to a saturation of ideas (80 unique design ideas). In the last two sessions, for example, the vast majority of ideas were already generated previously. The saturation was also evident in the subsequent focus group, which did not generate any new ideas. Overall, this may have to do with the fact that the participants all had similar professional backgrounds.

The ideas were grouped into themes such as play, family, social, communication, safety, eco/efficiency, points of interest, civic engagement, education, self-organisation/management and entertainment. Some of them have previously been discussed [8, 7].

The subsequent evaluation also exposed interesting and unexpected issues, as well as opportunities. A detailed account of the results was presented in [9], we summarize here some critical aspects that evidence the usefulness of

the methodological approach.

The risk of distraction was perceived differently by participants depending on age, gender and habitual driving behaviour. Reasons may depend on a overconfidence or lack of experience of younger drivers, in a different and more open attitude towards technology in general, or in an actual difference of cognitive load for different users.

Participants also provided unexpected insights into possible collateral effects, besides distraction, of the proposed technologies, repeatedly suggesting the risk (or even the intention) to use some of the applications to verbally abuse other drivers, or to be more aggressive in response to other drivers' use of the application.

All in all the methodology allowed the participants to effectively imagine or re-evolve the feelings, goals and behaviours of a real driving condition. The result was an abundance of design ideas, comments and insights that will inform our future research.

Acknowledgements

This research is supported under the Australian Research Council's Early Career Researcher Award (DECRA) funding scheme (project number DE140101542).

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